



Activities Terminal May 2 12:04 PM meetha156@localhost:~ /usr/bin/vim producer\_consumer.c

```
void *consumer(void *arg) {
    int item;

    sem_wait(&full);
    sem_wait(&mutex);

    item = buffer[out];
    printf("Consumer consumes item %d\n", item);

    out = (out + 1) % SIZE;

    sem_post(&mutex);
    sem_post(&empty);

    pthread_exit(NULL);
}

int main() {
    int choice;
    pthread_t p, c;

    sem_init(&empty, 0, SIZE);
    sem_init(&full, 0, 0);
    sem_init(&mutex, 0, 1);

    while (1) {
        printf("\n1. Producer\n2. Consumer\n3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                if (sem_trywait(&empty) == 0) {
                    sem_post(&empty); // restore
                    pthread_create(&p, NULL, producer, NULL);
                    pthread_join(p, NULL);
                } else {
                    printf("Buffer is full!!\n");
                }
                break;
        }
    }
}
```

36,0-1 60%

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```
sem_init(&empty, 0, SIZE);
sem_init(&full, 0, 0);
sem_init(&mutex, 0, 1);

while (1) {
    printf("\n1. Producer\n2. Consumer\n3. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            if (sem_trywait(&empty) == 0) {
                sem_post(&empty); // restore
                pthread_create(&p, NULL, producer, NULL);
                pthread_join(p, NULL);
            } else {
                printf("Buffer is full!!\n");
            }
            break;

        case 2:
            if (sem_trywait(&full) == 0) {
                sem_post(&full); // restore
                pthread_create(&c, NULL, consumer, NULL);
                pthread_join(c, NULL);
            } else {
                printf("Buffer is empty!!\n");
            }
            break;

        case 3:
            exit(0);

        default:
            printf("Invalid choice\n");
    }
}

return 0;
}
```

91,1 Bot

```
Activities Terminal May 2 12:15 PM meetha156@localhost:~  
[meetha156@localhost ~]$ vi producer_consumer.c  
[meetha156@localhost ~]$ gcc producer_consumer.c -o pc -lpthread  
[meetha156@localhost ~]$ ./pc  
1. Producer  
2. Consumer  
3. Exit  
Enter your choice: 1  
Producer produces the item 83  
1. Producer  
2. Consumer  
3. Exit  
Enter your choice: 2  
Consumer consumes item 83  
1. Producer  
2. Consumer  
3. Exit  
Enter your choice: 2  
Buffer is empty!!  
1. Producer  
2. Consumer  
3. Exit  
Enter your choice: 1  
Producer produces the item 86  
1. Producer  
2. Consumer  
3. Exit  
Enter your choice: 1  
Producer produces the item 77  
1. Producer  
2. Consumer  
3. Exit  
Enter your choice: 1  
Producer produces the item 15  
1. Producer  
2. Consumer
```